TENNESSEE VALLEY AUTHORITY

Post Office Box 2000, Decatur, Alabama 35609-2000

January 10, 2025

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Unit 1
Renewed Facility Operating License No. DPR-33
NRC Docket No. 50-259

Subject: **Licensee Event Report 50-259/2024-004-00 – Main Steam Relief Valves Lift Settings Outside of Technical Specifications Required Setpoints**

The enclosed Licensee Event Report provides details of Main Steam Relief Valves which failed to meet their Surveillance Requirements for longer than allowed by plant Technical Specifications. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's Technical Specifications.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact David J. Renn, Nuclear Site Licensing Manager, at (256) 729-2636.

Respectfully,

DaK

Daniel A. Komm
Site Vice President

Enclosure: Licensee Event Report 50-259/2024-004-00 – Main Steam Relief Valves Lift Settings Outside of Technical Specifications Required Setpoints cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant
NRC Project Manager - Browns Ferry Nuclear Plant

NRC FORM 366 (04-02-2024) U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Browns Ferry Nuclear Plant, Unit 1	☒ 050 ☐ 052	00259	1 OF 6

4. Title
Main Steam Relief Valves Lift Settings Outside of Technical Specifications Required Setpoints

5. Event Date			6. LER Number			7. Report Date		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year
11	12	2024	2024	- 004 -	00	01	10	2025

8. Other Facilities Involved	
Facility Name	Docket Number
N/A	005000 N/A
N/A	050000 N/A

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Ryan Coons, Licensing Engineer	256-729-2070

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	SB	RV	T020	N	N/A	N/A	N/A	N/A	N/A

4. Supplemental Report Expected)		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date		N/A	N/A	N/A

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On November 12, 2024, the Tennessee Valley Authority (TVA) was notified of as-found testing results that two (2) Main Steam Relief Valves (MSRVs) from Browns Ferry Nuclear Pant (BFN), Unit 1 were outside of their Surveillance Requirement limits for an indeterminate amount of time. These affected valves remained capable of maintaining reactor pressure within the American Society of Mechanical Engineers code limits and were still able to perform their safety function.

It was determined that one MSRV failed above its setpoint due to corrosion bonding to the valve seats. The other MSRV failed below its setpoint due to relaxed setpoint springs. All thirteen of the MSRV pilot valves were replaced during the Unit 1 Fall 2024 refueling outage. The Boiling Water Reactor Owners' Group is continuing to work toward a solution to improve the quality and adhesion of the platinum coating on the pilot discs. BFN will continue to verify that the springs pass spring rate testing during their rebuilding and replacing those which do not.

NARRATIVE

I. Plant Operating Conditions before the Event

At the time of discovery, Browns Ferry Nuclear Plant (BFN) Unit 1 was in Mode 1 at approximately 100 percent power.

II. Description of Event

A. Event Summary

On November 12, 2024, NTS/Element notified TVA with the as-found testing results of the thirteen Main Steam Relief Valves (MSRVs) [RV], which were removed during the Fall 2024 Unit 1 Refueling Outage 15 (U1R15). Two (2) MSRVs (1-PCV-001-0005 and 1-PCV-001-0180) had as-found lift settings which were outside of the +/- 3 percent band of their setpoints required by Technical Specification (TS) 3.4.3, Safety/Relief Valves (S/RVs) for an indeterminate amount of time.

Throughout this event, the two stage MSRV pilot valves remained capable of maintaining reactor pressure below 1375 pounds per square inch gauge (psig), which is the American Society of Mechanical Engineers (ASME) code limit of 110 percent of the vessel design pressure. The valves remained capable of performing their required safety function.

TVA is submitting this report in accordance with Title 10 of the Code of Federal Regulations 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's TS.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

There were no structures, systems, or components (SSCs) whose inoperability contributed to this event.

C. Dates and approximate times of occurrences

Dates and Approximate Times	Occurrence
October 26, 2022	Unit 1 entered Mode 2, beginning Fuel Cycle 15 (U1C15).
August 30, 2024	Unit 1 OPS inserted Manual Reactor SCRAM to begin U1R15.
November 12, 2024	NTS/Element notified TVA with as-found testing results of the thirteen Unit 1 MSRV pilot valves removed during U1R15.

NARRATIVE

D. Manufacturer and model number of each component that failed during the event

The failed components were all Target Rock Corporation two stage pressure control valves, model number 7567F.

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

F. Method of discovery of each component or system failure or procedural error

The MSRV failures were discovered at NTS/Element during as-found testing of the thirteen MSRV two-stage pilot valves which were removed during U1R15.

G. The failure mode, mechanism, and effect of each failed component

MSRV 1-PCV-001-0005 failed above its setpoint due to corrosion bonding to the valve seat and a non-quantifiable leakage known as simmering.

MSRV 1-PCV-001-0180 failed below its setpoint due to a relaxation of the setpoint spring over time.

H. Operator actions

There were no operator actions associated with this event.

I. Automatically and manually initiated safety system responses

There were no automatic or manual safety system responses associated with this event.

III. Cause of the event

A. Cause of each component or system failure or personnel error

MSRV 1-PCV-001-0005 failed above its setpoint due to corrosion bonding to the valve seat and a non-quantifiable leakage known as simmering. As a result of corrosion bonding, the force required to break the crystal structure of the corrosion bond alters the mechanical setpoint of the pilot valve. This issue is commonly known to the industry as set point drift. Simmering valves result in mechanical setpoint drift and is generally the result of low stellite in the seat.

MSRV 1-PCV-001-0180 lifted below its setpoint due to relaxed springs.

NARRATIVE

B. Cause(s) and circumstances for each human performance related root cause

No human performance related root causes were identified.

IV. Analysis of the event

BFN, Unit 1 TS Surveillance Requirement (SR) 3.4.3.1 specifies verification of lift settings of the required 12 Safety/Relief Valves (S/RVs) are within a +/- 3 percent band of their setpoint values in accordance with the Inservice Testing Program. BFN, Unit 1 has thirteen MSRVs to satisfy this requirement with margin. When tested, the following S/RVs were outside the allowable +/- 3 percent band:

S/RV ID Number	Setpoint (psig)	Test Result (psig)	Difference (Percent)
1-PCV-001-0005	1145	1227.0	+7.16
1-PCV-001-0180	1155	1114.6	-3.50

Prior to startup from U1R15 all thirteen MSRV pilot valves were replaced with refurbished valves which were certified to lift within +/- 1 percent of their setpoint. Operating Experience has shown that Target Rock two stage MSRV setpoint drift is not a uniform, linear process. The corrosion bonding increases at a random rate.

TS LCO 3.0.4 states that when an LCO is not met, entry into a Mode or other specified condition in the Applicability shall only be made when the associated actions to be entered permit continued operation in the Mode or other specified condition in the Applicability for an unlimited period of time. BFN, Unit 1 entered TS 3.4.3 Applicable Modes when LCO TS 3.4.3 Required Actions may not have been met on the following dates:

- February 1, 2023
- May 22, 2023
- January 30, 2024
- May 5, 2024

V. Assessment of Safety Consequences

System availability was not impacted by this event. The failure of the MSRV pilot valves to meet their TS 3.4.3 specified mechanical setpoints did not impact their remote manual operation or activation through the MSRV Automatic Actuation Logic, because these operating modes and functions rely upon electrically signaled control air solenoids to open the MSRV pilot valves.

TS Bases 3.4.3 states that the overpressure protection system must accommodate the most severe pressurization transient. The MSRVs remained capable of maintaining the reactor

NARRATIVE

pressure below 1375 psig, which is the ASME code limit (110 percent of the vessel design pressure). The valves remained capable of performing their required safety function.

Based on the above, the TVA has concluded that sufficient systems were available to provide the required safety functions needed to protect the health and safety of the public.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

Each BFN operating unit has a non-safety related, electrical logic system (MSRV Actuation Logic) installed, which provides defense in depth against MSRV setpoint drift by electrically opening MSRV groups based upon setpoints at 1135 psig, 1145 psig, and 1155 psig. Therefore, during a reactor pressure transient event, the four 1135 psig group MSRVs, followed by the four 1145 psig group MSRVs, and finally the five 1155 psig group MSRVs would receive an electrical open signal, providing a defense in depth function to allow the valves to perform their safety function.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when the reactor was shutdown.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

TS 3.4.3 requires twelve of the thirteen S/RVs to be operable for S/RV system operability. While two of thirteen valves failed to meet their TS SR 3.4.3.1 limits for an indeterminate period, they remained capable of performing their required safety function.

VI. Corrective Actions

Corrective Actions are being managed by the TVA's corrective action program under Condition Reports (CRs), 962223, 1286467, 1410577, 1521190, 1658693, 1699286, 1775232, 1822254, 1860559, 1929514, and 1972030.

A. Immediate Corrective Actions

All thirteen of the BFN, Unit 1 MSRV pilot valves were replaced with refurbished valves during U1R15. As left testing verified that these refurbished pilot valves were within +/- 1 percent of their setpoints.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
Browns Ferry Nuclear Plant, Unit 1	☒ 050 / ☐ 052	00259	2024	- 004	- 00

NARRATIVE

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

As discussed in LER 50-260/2021-002-00, a flaking issue has been noted with the platinum coated pilot discs. The Boiling Water Reactor Owners' Group (BWROG) is continuing to work toward a solution to improve the quality and adhesion of the platinum coating on the discs. The corrective actions suggested by the BWROG will be incorporated to correct setpoint drift. To reduce the probability of seat leakage occurring in the future, pilot seat rebuilds will be performed in valves that have low stellite in the seat.

BFN will take additional actions to verify that the springs associated with each low lliftpoint passes Spring Rate testing or is replaced.

VII. Previous Similar Events at the Same Site

A search of LERs from BFN, Units 1, 2, and 3 over the last five years identified seven LERs associated with MSRV lift settings outside of TS required setpoints:

- LER 50-296/2024-003-00, for Unit 3 Cycle 21
- LER 50-260/2023-002-00, for Unit 2 Cycle 22
- LER 50-259/2022-003-00, for Unit 1 Cycle 14
- LER 50-296/2022-001-00, for Unit 3 Cycle 20
- LER 50-260/2021-002-00, for Unit 2 Cycle 21
- LER 50-259/2020-003-01, for Unit 1 Cycle 13
- LER 50-296/2020-002-00, for Unit 3 Cycle 19

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.